

Mapping direct anthropogenic drivers of biodiversity change using large language models: a systematic map protocol

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1 Background

2 Achieving the 2050 Vision of the Kunming-Montreal Global Biodiversity Framework re-
3 quires understanding what drives biodiversity loss across all drivers, ecosystems, and ge-
4 ographies [1]. Direct anthropogenic drivers act immediately on ecosystems, species, and
5 populations [1, 2], yet the scientific evidence base on these drivers remains uneven. A
6 small subset of taxa receives the majority of research attention [3], and the same imbal-
7 ance holds across drivers: climate change accounts for nearly 40% of driver-related pub-
8 lications, while pollution is represented in fewer than 5% [4]. Most evidence syntheses
9 address a single driver, taxon, or region [5–9], leaving the broader evidence base across
10 drivers largely uncharacterized.

11 This systematic map will apply a broad search strategy for biodiversity change and
12 direct anthropogenic drivers to screen over 2.4 million candidate bibliographic records.
13 Following the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem
14 Services (IPBES) conceptual framework [2], eligible records will be coded to produce a
15 structured evidence base across drivers, threats, geography, ecosystems, study attributes,
16 and taxa. The rapidly expanding biodiversity literature makes comprehensive manual syn-
17 thesis increasingly challenging [10]; therefore, this map will use Large Language Models
18 (LLMs) to automate screening and coding within the Collaboration for Environmental Ev-
19 idence (CEE) systematic mapping framework. The map will help biodiversity researchers
20 and policymakers identify evidence gaps and research priorities across direct anthropo-
21 genic drivers.

22 **Theory of Change or Causal Model**

23 This systematic map does not evaluate interventions and does not construct a causal model.
24 Following the IPBES conceptual framework [2], it characterizes the scientific evidence
25 base on direct anthropogenic drivers of biodiversity change. The map draws on this
26 framework to define the exposure (direct anthropogenic drivers) and outcome (biodiversity
27 change), as illustrated in Supplementary Figure 1.

28 **Stakeholder Engagement**

29 No formal external stakeholder consultation will be conducted. However, screening and
30 coding outputs will be compared against labels from previously published sources on bio-
31 diversity change through the consistency checking procedures described in this protocol.

32 **Objectives and Review Question**

33 The objective of this systematic map is to collect, code, and characterize the scientific ev-
34 idence on the direct anthropogenic drivers of biodiversity change, with primary analytical
35 focus on biodiversity loss. The map is guided by two overarching questions:

- 36 1. What scientific evidence is available on the direct anthropogenic drivers of biodiver-
37 sity change?
- 38 2. How is this evidence characterized across direct anthropogenic drivers, specific
39 threats, geography, ecosystems, study attributes, and taxa?

Definitions of the Question Components

The review questions are defined using the Population, Exposure, and Outcome (PEO) framework.

Population: This map broadly covers total biodiversity, the variability among living organisms from all sources [11]. No geographic or taxonomic restriction is applied; all taxonomic groups, ecosystems, and geographies will be coded using pre-defined classification schemes [2, 12–15].

Exposure: The exposure comprises any direct anthropogenic driver acting on biodiversity, classified using the five IPBES driver categories [2] and further resolved using the International Union for Conservation of Nature (IUCN) Threats Classification [12].

Outcome: The outcome is reported biodiversity change at genetic, species, community, or ecosystem level. The primary focus is negative change.

Search Strategy

The literature search will be conducted on Web of Science Core Collection (WoS), with publication years 2000–2026; records with publication-date metadata after 30 April 2026 will be excluded at the point of export. The search string is designed to prioritize recall over precision, capturing a broad corpus (see Table 1). It comprises five components combined using AND: a publication-type filter (A); a scientific disciplines filter via WoS subject categories (B); a change/assessment terms block (C); five driver-specific sub-strings, one per IPBES direct driver category, combined using OR and adopted from a prior rapid evidence assessment [6] (D); and a publication-year filter (E).

Component	Operator	What this block does	WoS field	Search string
(A) Publications	AND	Limit results to articles only.	DT	DT=(Article)
(B) Scientific disciplines	AND	Limits results to relevant Web of Science Categories to keep retrieval manageable.	WC	WC=("Biodiversity Conservation" OR "Ecology" OR "Environmental Sciences" OR "Environmental Studies" OR "Marine & Freshwater Biology" OR "Fisheries" OR "Oceanography" OR "Forestry" OR "Water Resources" OR "Agronomy" OR "Geosciences, Multidisciplinary" OR "Meteorology & Atmospheric Sciences" OR "Plant Sciences" OR "Zoology" OR "Remote Sensing" OR "Multidisciplinary Sciences")
(C) Change / assessment / monitoring terms	AND	Ensures retrieved records include language indicating measured change or assessment/monitoring (e.g., loss/change, impacts/pressures, trends/responses, risk, monitoring).	TS	TS=((loss* NEAR/3 (habitat* OR biodiversit* OR species OR population* OR "ecosystem service*" OR ecosystem* OR forest* OR coral* OR mangrove* OR reef* OR wetland* OR bird* OR mammal* OR plant*)) OR change* OR reference* OR indicator* OR impact* OR effect* OR pressure* OR trend* OR response* OR status OR pattern* OR assessment* OR monitor* OR risk* OR driver*)
(D) Direct drivers of anthropogenic biodiversity change (driver sub-strings combined with OR within TS)				
(D1) Land/sea use change	OR	Terms capturing land/sea use change and habitat modification.	TS	((("land use chang*" OR "land cover chang*" OR "land* system* chang*" OR "land* chang*" OR "land* degradation" OR "land* conversion" OR "land management") OR ((marine OR ocean OR sea*) AND use AND change*) OR ("habitat loss*" OR "habitat degradation*" OR "habitat chang*" OR "habitat conversion" OR "habitat modification*" OR "habitat fragmentation" OR "habitat management") OR (biofuel* OR biogas*) OR ((residential OR urban* OR cities OR housing OR commercial OR industrial OR tourism* OR recreation*) AND (extent OR area OR cover* OR proportion* OR percentage*) AND (develop* OR expand* OR increase*)) OR (deforestation OR afforestation OR "wood plantation*" OR "forest* plantation*" OR "pulp* plantation*") OR ((aquaculture* OR agricultur* OR crop* OR "cultivated area*" OR farmland OR pasture) AND (develop* OR expand* OR increase* OR intensification OR extensification OR abandon* OR change*)) OR (livestock AND (farm* OR ranch* OR graz* OR densit*)) OR (energy AND (drill* OR min* OR quarry*)) OR (((transport* OR service) AND (network* OR corridor* OR infrastructure*)) OR road* OR rail* OR "ship* lane*" OR "flight path*") OR ("system* modification*" OR (dams OR (water AND (management OR use))))))
(D2) Direct Exploitation and Resource Extraction	OR	Terms capturing extraction/harvest/withdrawal of biological and abiotic resources, including fishing and logging.	TS	(((((natur* AND resource*) OR material* OR (biomass OR "fossil fuel*" OR ore* OR mineral*)) OR ((biological AND resource*) OR wildlife OR animal* OR plant* OR population*) OR water) AND ((overextraction OR overexploitation) OR (extraction OR exploitation OR withdraw*) OR depletion OR (hunting OR collect* OR gather* OR poaching OR "bush meat*"))) OR ((forest* OR wood OR timber*) AND (harvest* OR logging*)) OR (soil* AND (erosion OR acidification OR degradation)) OR (fisher* OR fishing OR (bycatch OR by-catch))
(D3) Climate change	OR	Terms capturing climate change and associated climatic stressors and impacts.	TS	("climat* change" OR "chang* climat*" OR ("increase* temperature*" OR "thermal alteration" OR "global warming") OR (chang* AND precipitation*) OR ((extreme OR severe) AND (weather OR temperature OR precipitation OR climat*)) OR heatwave* OR drought* OR storm* OR flood* OR (UV OR ultraviolet) OR "sea level rise" OR "soil salinization" OR ("habitat shift*" AND climate) OR ("habitat alteration" AND climate) OR ((consumptive OR human OR agricultural) AND "water stress"))
(D4) Pollution	OR	Terms capturing pollution and contaminants across environmental media (soil/water/air).	TS	((pollut* OR contamin* OR emission* OR spill-over* OR disposal* OR deposition* OR load* OR dump* OR discharge*) OR ((noise OR light OR gas* OR particle* OR particulate* OR nitrogen OR sulphur* OR phosphor* OR waste* OR garbage OR sewage OR pesticide* OR fertilizer* OR nutrient* OR "oil spill*" OR metal OR salinization OR salinisation OR acidification OR solid OR plastic OR mercury OR nutrient* OR pollutant*) AND (soil OR water OR ocean OR marine OR atmosphere OR air)) OR (((industrial OR military OR agricultur* OR forest*) AND effluent*))
(D5) Invasive alien species	OR	Terms capturing invasive alien species introductions and biological invasions.	TS	((((invasive OR invasion) AND (alien OR exotic OR non-native OR pest* OR disease) AND species) OR "biological invasion*" OR (introduc* AND (species OR material)))
(E) Publication year	AND	Restricts results to the selected publication year range.	PY	PY=2000-2026

Table 1: Search string components used in Web of Science: Components (A)–(E) are combined using AND. Within TS, the five direct driver sub-strings (D1–D5) from [6] are combined using OR. Component (E) applies the WoS publication-year filter PY=2000–2026; because WoS publication-year filtering does not express a day/month cutoff, records with publication-date metadata after 30 April 2026 will be excluded at the point of export. A preliminary feasibility check conducted on 02 May 2026 yielded over 2.4 million unique records after deduplication, confirming the scale of the corpus to be screened and coded.

61 **Bibliographic Databases**

62 The search will be executed on the WoS Core Collection. Throughout the search, no lan-
63 guage restrictions will be applied. Gray literature is excluded in this systematic map; pro-
64 cessing of gray literature at this corpus scale falls outside the scope of the LLM pipeline.
65 This is acknowledged as a limitation of the search coverage.

66 **Web-Based Search Engines**

67 No web-based search engines will be searched.

68 **Organizational Websites**

69 No organizational websites will be searched.

70 **Comprehensiveness of the Search**

71 Comprehensiveness was assessed against an *a priori* test list of 581 publications com-
72 piled from five published sources (three meta-analyses, a rapid evidence assessment, and
73 a systematic map protocol) on direct drivers of biodiversity loss [5–9] and the IPBES
74 Global Assessment chapters on direct drivers (sections 2.1.12–2.1.17) [16]. Together,
75 these sources span all five IPBES direct driver categories.

76 The test list was assembled by extracting reference lists from each source and exclud-
77 ing gray literature and records not indexed in WoS, yielding 581 records from an initial 791
78 candidates. The final search string retrieved 497 of these records, yielding source-specific
79 recall ranging from 71.43% to 97.32% and an overall recall of 85.54%.

80 **Search Update**

81 No search updates are planned.

82 **Screening Strategy**

83 The literature search will retrieve over 2.4 million bibliographic records spanning all five
84 direct anthropogenic drivers of biodiversity change. All eligibility screening decisions will
85 be made by GPT-5-nano (version 2025-08-07). Prompts and the reasoning effort pa-
86 rameter will be finalized on the training and development splits of the validation datasets
87 and frozen prior to full-corpus application; prompts tuning and freezing procedure is de-
88 scribed in Supplementary Text S2.

89 The model will apply a four-question procedure (Q1–Q4) to the title and abstract of
90 each record; each question returns a structured label alongside a result of `Yes`, `Unclear`,
91 or `No`. Screening will proceed in three stages:

- 92 1. *Deduplication*: Duplicate records will be removed from the raw retrieved corpus using
93 the WoS primary key of each bibliographic record.
- 94 2. *Abstract availability*: Records without an abstract will be excluded, as the abstract is
95 the primary source of text for screening and data coding.
- 96 3. *Eligibility screening (Q1–Q4)*: Each remaining record will be screened using the four-
97 question procedure. A record will be classified as eligible only when all four questions
98 return `Yes` or `Unclear`; a single `No` at any question yields an ineligible classification.
99 `Unclear` responses will be retained to avoid false exclusions.

100 This map applies no full-text screening stage, deviating from CEE guidelines. This en-
101 ables consistent and reproducible screening at a scale infeasible for traditional systematic
102 maps.

103 **Eligibility Criteria**

104 Each of the four screening questions targets one component of the evidence definition (see
105 Figure 1 and Supplementary Text S1). A record will be excluded only when at least one
106 question returns No.

- 107 • *Biodiversity change*: Does the record report at least one form of biodiversity
108 change? Change is operationalized across four forms: genetic change, species
109 change, community change, and ecosystem change.
- 110 • *Direction of change*: Does the record identify a direction of biodiversity change?
111 Direction is recorded as negative, positive, mixed, or unclear. The primary analytical
112 focus is negative change, i.e., loss of biodiversity.
- 113 • *Direct anthropogenic driver*: Does the record mention or plausibly imply at least
114 one direct anthropogenic driver? Drivers are assessed using the five IPBES driver
115 categories [2] and the IUCN Threats Classification Scheme [12].
- 116 • *Linkage*: Does the record report or plausibly imply a link between the identified
117 driver and the biodiversity change?

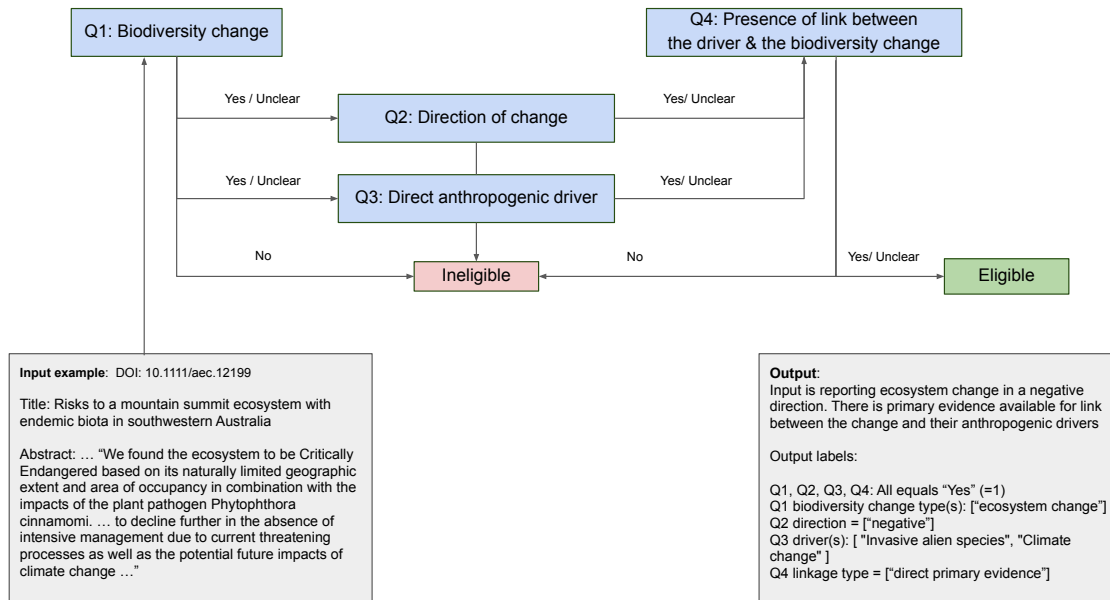


Figure 1: **Eligibility screening procedure:** Each record is screened using four questions (Q1–Q4) applied to its title and abstract. A record is classified as eligible only when all four questions return Unclear or Yes; a single No at any question yields an ineligible classification. Each question produces a structured label alongside its result. See Supplementary Table 2 for additional examples.

118 Consistency Checking

119 Eligibility screening deviates from CEE guidelines in two respects: all decisions will be
 120 made by GPT-5-nano rather than multiple human reviewers, and no full-text screening
 121 stage will be applied. Both are acknowledged as limitations; their effect on screening
 122 reliability will be assessed through two consistency checks against expert full-text labels,
 123 defined as eligibility decisions assigned by the original authors of the reference sources
 124 based on full-text reading (see Table 2). As each reference source applies its own inclusion
 125 criteria, expert labels may not map perfectly with the eligibility criteria of this map; the

alignment between these criteria is treated as a working assumption of the consistency checking procedure.

Consistency Check 1 (CC1) will compare author-assigned labels from abstracts only against expert full-text and LLM labels, establishing a human recall baseline; Consistency Check 2 (CC2) will evaluate LLM labels against expert full-text labels. As screening will be performed by GPT-5-nano, no inter-rater disagreement will arise; reliability will be assessed using recall as the primary metric to minimize false exclusions at the screening step.

Dataset	Size	Details	Role
Reference Screening Dataset	$n = 2,418$	Records included in three meta-analyses [17–19], a rapid evidence assessment [6], and the Red List of Ecosystems Database [20].	CC2: LLM labels evaluated against expert full-text labels to assess screening recall
Annotated Screening Dataset	$n = 500$	Author-assigned screening labels based on abstracts only, samples drawn from the Reference Screening Dataset and search query results. No full-text access allowed during annotation.	CC1: author-assigned labels from abstracts compared against expert full-text and LLM labels, establishing a human recall baseline

Table 2: Validation datasets supporting the screening consistency checks.

Reporting Screening Outcomes

Screening outcomes will be reported in full to support transparency and reproducibility. The following will be provided:

1. A Reporting Standards for Systematic Evidence Syntheses (ROSES) flow diagram documenting record counts at each stage of the screening pipeline, from raw retrieval to final eligible records.
2. The raw literature retrieval dataset exported from WoS in Parquet format.

- 141 3. The full screening output dataset in Parquet format. The field structure is defined in
142 Supplementary Table 1.
- 143 4. The LLM prompt, model configuration, and raw model output at each eligibility stage
144 for each record.

145 **Study Validity Assessment**

146 Under CEE guidelines, critical appraisal of study validity is optional for systematic maps
147 and does not influence the mapping process itself. No formal risk-of-bias appraisal will be
148 conducted for exclusion or weighting of included studies. Instead, study validity indicators
149 will be captured through the coding process: study design, data collection method, and
150 analysis method will be coded for every eligible record via the study attributes label set
151 (L5).

152 **Consistency Checking**

153 As no formal critical appraisal is conducted, no dedicated consistency checking for this
154 stage is required. Study validity indicators coded via the study attributes label set (L5) are
155 subject to the coding consistency checks described in the Data Coding Strategy section.

156 **Data Coding Strategy**

157 Eligible records will be coded across six label sets to construct an evidence chain: which
158 drivers are reported as linked to biodiversity change (L1), through which specific threat

categories (L2), in which locations and ecosystems (L3; L4), characterized by which study attributes (L5), and affecting which taxa (L6).

All coding decisions will be made by GPT-5-nano (version 2025-08-07). The model will code each record by applying pre-defined decision rules to its title and abstract. Prompts will use zero-shot instructions, refined iteratively on training and development splits prior to full-corpus application (see Supplementary Text S2 for prompts tuning and freezing procedure).

Where applicable, label sets will draw on controlled vocabularies from established classification schemes: the IPBES Direct Drivers, the IUCN Threats Classification Scheme, IPBES Geographic Regions, the Global Ecosystem Typology (GET), and the GBIF Backbone Taxonomy [2, 12–15].

Two coding modes will be applied depending on label set structure. Flat-structured label sets (L1, L3, L5) will be coded in a single prompt call per record. Hierarchical label sets (L2, L4) will be coded in staged multiple-pass sequences, where each level's output constrains candidate labels at the next level, avoiding inconsistent assignments. The taxonomy label set (L6) will be applied conditionally: only records flagged by L5 as taxon-focused will enter the taxonomy coding step.

Metadata to be Coded

Six label sets will be coded for each eligible record (see output schema in Supplementary Table 3).

IPBES Direct Drivers (L1): broad direct anthropogenic driver category as per the IPBES framework [2].

181 *IUCN Threats* ($L2$): specific proximate threats to biodiversity, resolved across three
182 hierarchical levels [12].

183 *Geographic Scope* ($L3$): IPBES region, sub-region, and country [13], together with
184 named sub-national locales and coordinates where reported or inferred from locale names.

185 *Ecosystem Typology* ($L4$): Global Ecosystem Typology (GET) realm, biome, and Eco-
186 system Functional Group [14].

187 *Study Attributes* ($L5$): study design, methods of data collection and analysis, compara-
188 tor presence and type where reported, and a taxon-focus flag.

189 *Taxonomy* ($L6$): taxonomic rank, and scope from kingdom to species, resolved against
190 the GBIF Backbone Taxonomy [15], applied conditionally to records where $L5$ returns a
191 taxon-focus flag.

192 **Consistency Checking**

193 Data coding deviates from CEE guidelines in the same two respects as eligibility screen-
194 ing; their effect on coding reliability will be assessed through two consistency checks
195 against expert full-text labels, defined as coded labels assigned by the original authors of
196 the reference sources based on full-text reading (see Table 3).

197 Consistency Check 1 (CC1) will compare author-assigned labels from abstracts only
198 against expert full-text labels, establishing a human-performance baseline for each label
199 set; Consistency Check 2 (CC2) will evaluate LLM labels against expert full-text labels
200 across all six label sets ($L1$ – $L6$). As coding will be performed by GPT-5-nano, no inter-
201 rater disagreement will arise. Reliability will instead be assessed using precision, recall,
202 F_1 , and Cohen’s κ .

Dataset	Size	Details	Role
Reference Coding Dataset	$n = 7,844$	Records included in three meta-analyses [8, 9, 18], two systematic reviews [21, 22], two systematic maps [23, 24], a rapid evidence assessment [6], and one systematic review preprint [25], carrying expert full-text labels across applicable label sets L1–L6.	CC2: LLM labels evaluated against expert full-text labels across all six label sets (L1–L6)
Annotated Coding Dataset	$n = 600$ (100 per label set)	Author-assigned coding labels based on abstracts only, samples drawn from the Reference Coding Dataset. No full-text access allowed during annotation.	CC1: labels assigned from abstracts compared against expert full-text labels across all six label sets (L1–L6)

Table 3: Validation datasets supporting the data coding consistency checks.

Type of Mapping

This systematic map will be produced as a coded tabular dataset in which each row represents one eligible record enriched with labels from all six coding dimensions (L1–L6). Because the pipeline is fully automated and reproducible, the map is designed as a *living systematic map*: it can be re-run against new literature without manual screening and coding.

The dataset will combine a full bibliographic record export from WoS with the meta-data fields defined in Supplementary Table 3. Full screening outputs will be provided as a separate dataset; the field structure is defined in Supplementary Table 1. Both will be made available in a publicly accessible repository alongside the map. The coded dataset will be accompanied by an interactive web application for evidence exploration.

214 **Narrative Synthesis Methods**

215 Narrative synthesis will characterize the distribution of evidence across all six coded di-
216 mensions through descriptive statistics on counts and proportions per category. Cross-
217 tabulations of driver by geography, driver by ecosystem, and driver by study attributes will
218 be reported as matrices. Temporal trends in publication volume by driver category will
219 document how the evidence base has evolved since 2000. No effect sizes are estimated
220 and no causal conclusions are drawn.

221 **Knowledge Gap Identification Strategy**

222 Knowledge gaps will be identified by inspecting cross-tabulation matrices for sparse or
223 absent driver-dimension combinations across geography, ecosystems, study attributes, and
224 taxa.

225 **Demonstrating Procedural Independence**

226 Eligibility screening and data coding will be performed by an LLM pipeline that applies
227 fixed prompts and decision criteria uniformly to all records. Any records authored by
228 review team members will be processed by the same frozen LLM pipeline as all other
229 records; no reviewer discretion will be applied. All pipeline code, prompts, and decision
230 rules will be documented in the supplementary code repository.

231 **Competing Interests**

232 The authors declare no competing interests.

233 **Funding Information**

234 No external funding was received.

235 **Authors' Contributions**

236 H.M. developed the protocol and wrote the manuscript. K.F. reviewed the manuscript and
237 provided feedback. S.F. conceived the study, provided supervisory guidance, and reviewed
238 the manuscript. All authors read and approved the final protocol manuscript.

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240 We thank the authors of the published evidence syntheses used to construct the search test
241 list and the validation datasets for screening and coding.

242 **Data Availability**

243 The data repository contains the screening and coding validation datasets used to support
244 the consistency checking procedures, LLM prompts for all label sets, and predefined label

²⁴⁵ classifications with source definitions for each label set. All materials are available at:
²⁴⁶ <https://osf.io/sna2g/>.

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308 **Supplementary Tables and Figures**

Field	Type	Description
<i>Record identifiers and processing metadata</i>		
custom_id	String	Unique identifier for the record in the original literature retrieval.
_source_run	String	Identifier for the processing partition in which the record was screened.
model	String	Name of the LLM used to generate the screening output.
created_at	Datetime	Timestamp indicating when the record was processed.
raw_output	String	Original LLM response before parsing into structured fields.
<i>Screening outputs Q1–Q4</i>		
q1_r	Integer	Q1 result: biodiversity change reported? 1 = Yes, −1 = Unclear, 0 = No.
q1_label	List[String]	Biodiversity change type(s): genetic change, species change, community change, or ecosystem change.
q2_r	Integer	Q2 result: direction identifiable? 1 = Yes, −1 = Unclear, 0 = No.
q2_label	List[String]	Direction label(s): negative, positive, mixed, or unclear.
q3_r	Integer	Q3 result: direct anthropogenic driver mentioned or plausibly implied? 1 = Yes, −1 = Unclear, 0 = No.
q3_label	List[String]	Driver or threat name(s) identified during screening; these support eligibility screening and are not the final coding outputs.
q4_r	Integer	Q4 result: driver–biodiversity link reported or plausibly implied? 1 = Yes, −1 = Unclear, 0 = No.
q4_label	List[String]	Linkage type(s) describing how the driver is linked to the reported biodiversity change.
<i>Eligibility decision</i>		
pred_screening	List[String]	Final eligibility decision derived from Q1–Q4: ["ELIGIBLE"] or ["INELIGIBLE"].
<i>Bibliographic details</i>		
Article Title	String	Title of the bibliographic record.
Abstract	String	Abstract of the bibliographic record.
Publication Year	Integer	Publication year of the bibliographic record.
Authors	String	Authors of the bibliographic record.
DOI	String	DOI of the bibliographic record, where available.

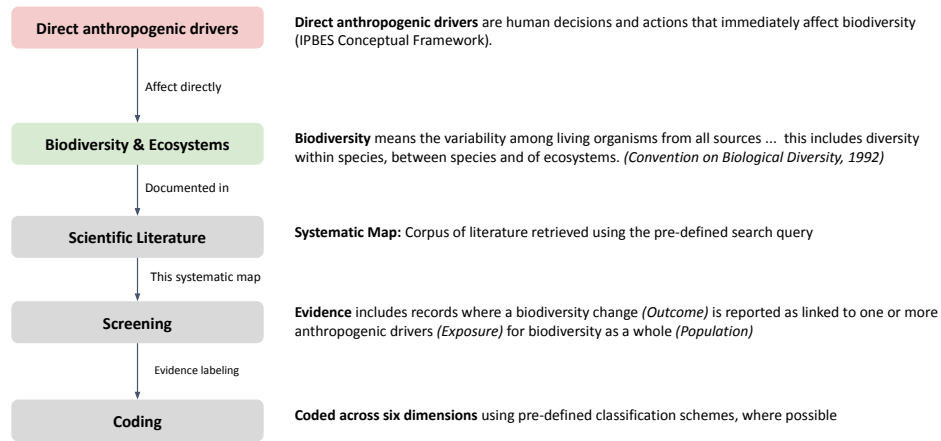
Supplementary Table 1: **Output fields recorded for each record in the screening dataset:** Screening fields (Q1–Q4) record the decision result and structured label(s) generated for each eligibility question.

#	Title	Abstract excerpt	Q1: Change	Q2: Dir.	Q3: Driver(s)	Q4: Linkage	Decision	Rationale
1	Risks to a mountain summit ecosystem with endemic biota in southwestern Australia	"... We found the ecosystem to be Critically Endangered based on its naturally limited geographic extent and area of occupancy in combination with the impacts of the plant pathogen <i>Phytophthora cinnamomi</i>"	Ecosystem change	Negative	Invasive alien species; Climate change	Direct primary evidence	Eligible	Reports ecosystem-level change in a negative direction, identifies two anthropogenic drivers, and provides direct empirical evidence linking them to the change.
2	Vulnerability of Antarctic shallow invertebrate-dominated ecosystems	"... Climate change is likely to cause early breakout of seasonal sea ice ... This is predicted to result in ecological regime shifts, in which invertebrate-dominated communities are replaced by macroalgal beds ..."	Ecosystem change; Community change	Negative	Climate change	Projected	Eligible	Reports both ecosystem- and community-level change, with regime shifts framed as negative. Linkage is projected because the driver–biodiversity connection is based on predicted future scenarios.
3	The effects of grazing and watering on ecosystem CO ₂ fluxes vary by community phenology	"... Grazing profoundly influences vegetation and the subsequent carbon fluxes in various ecosystems ... little effort has been made to explore the underlying mechanisms for phenological changes and their consequences on carbon fluxes at ecosystem level."	Ecosystem change	Both positive and negative directions identified	Grazing; Watering	Direct primary evidence	Eligible	The study examines anthropogenic grazing and watering effects on desert steppe and ecosystem carbon fluxes. The driver-change relationship is directly assessed through experiments.
4	Modeling Carbon Stocks in a Secondary Tropical Dry Forest in the Yucatan Peninsula, Mexico	"... The carbon balance of secondary dry tropical forests is sensitive to human and natural disturbances and climate change ... Carbon stocks were largely influenced by human disturbances between 1985 and 2010."	No change reported	No direction of change identified	Human disturbances; Climate change	No linkage present	Ineligible	Although anthropogenic drivers are identified, the study focuses on carbon stocks rather than biodiversity change.

Supplementary Table 2: Examples of the screening procedure.

Field	Type	Description
<i>L1: IPBES Direct Drivers</i>		
driver	List[String]	IPBES direct driver category or categories assigned to the record.
<i>L2: IUCN Threats</i>		
pred_threat_l0	List[String]	Broadest IUCN Threats Classification level assigned to the record.
pred_threat_l1	List[String]	Intermediate IUCN Threats Classification level assigned to the record.
pred_threat_l2	List[String]	Most specific IUCN Threats Classification level assigned to the record.
<i>L3: Geography</i>		
pred_regions	List[String]	IPBES region or regions of study focus.
pred_subregions	List[String]	IPBES sub-region or sub-regions of study focus.
pred_countries	List[String]	Countries of study focus, recorded using ISO 3166-1 alpha-3 codes.
locales	List[String]	Free-text names of sub-national locations mentioned in the title or abstract, such as parks, reserves, cities, or mountain ranges.
locale_coordinates	List[List[Float]]	[latitude, longitude] pairs in decimal degrees (WGS84, EPSG:4326); resolved from explicit coordinates in the title or abstract, or inferred from the locale name if absent.
<i>L4: Ecosystem Typology</i>		
pred_realm	List[String]	Global Ecosystem Typology realm or realms of study focus.
pred_biome	List[String]	Global Ecosystem Typology biome or biomes of study focus.
pred_efg	List[String]	Global Ecosystem Typology Ecosystem Functional Group or groups of study focus.
<i>L5: Study Attributes</i>		
pred_study_design	String	Study design category: Observational, Experimental, Modeling, or Unclear.
pred_methods_data_collection	List[String]	Data collection method or methods reported in the title or abstract, where applicable.
pred_methods_analysis	List[String]	Analysis method or methods reported in the title or abstract, where applicable.
pred_has_comparison	Boolean	True if the record explicitly states or strongly implies a comparator, such as pre/post, control versus exposed sites, or reference sites; otherwise false.
pred_comparison_types	List[String]	Type or types of comparison identified for the record, where applicable.
pred_taxa	Boolean	True if the record focuses on one or more specific taxonomic groups; otherwise false.
<i>L6: Taxonomy (conditional on pred_taxa = True)</i>		
rank	List[String]	Lowest taxonomic rank(s) identified for the focal taxon name(s) in the title or abstract.
kingdom	List[String]	Kingdom(s) resolved through the GBIF Backbone Taxonomy.
phylum	List[String]	Phylum/phyla resolved through the GBIF Backbone Taxonomy.
class	List[String]	Class(es) resolved through the GBIF Backbone Taxonomy.
order	List[String]	Order(s) resolved through the GBIF Backbone Taxonomy.
family	List[String]	Family/families resolved through the GBIF Backbone Taxonomy.
genus	List[String]	Genus/genera resolved through the GBIF Backbone Taxonomy.
species	List[String]	Species resolved through the GBIF Backbone Taxonomy.

Supplementary Table 3: Coded output fields for each eligible record across six label sets (L1–L6): L6 is applied conditionally to records where `pred_taxa = True` in L5. Where a field is not warranted by the record's scope, "Not Applicable" is coded; where a label cannot be determined from the title or abstract, "Unclear" is coded.



Supplementary Figure 1: **Conceptual Model:** Direct anthropogenic drivers are human decisions and actions that immediately affect biodiversity (IPBES Conceptual Framework). Scientific literature reporting biodiversity change linked to these drivers is retrieved via a systematic search, screened for eligibility, and coded across six dimensions using pre-defined classification schemes.

309 **Supplementary Text S1**

310 Biodiversity, as defined under the Convention on Biological Diversity (CBD) and op-
311 erationalized by IPBES, encompasses the variability among living organisms from all
312 sources, including terrestrial, marine, and other aquatic ecosystems, and the ecological
313 complexes of which they are part. This includes diversity within species, between species,
314 and of ecosystems [1, 11]. In this systematic map, eligibility is based on evidence of
315 biodiversity change, while the primary analytical focus is biodiversity loss. Evidence is
316 operationalized through four components: a reported biodiversity change, the direction of
317 that change, an anthropogenic driver linked to it, and the nature of the link between them.

318 **Biodiversity change:** Biodiversity change may be reported in one or more of four broad
319 forms:

- 320 1. *Genetic change*: such as change in genetic diversity or other heritable variation
321 within a species or population;
- 322 2. *Species change*: such as change in the abundance, occupancy, distribution, persis-
323 tence, or extinction risk of a taxon, species, or population;
- 324 3. *Community change*: such as change in richness, evenness, turnover, homogeniza-
325 tion, or composition of a species assemblage; and
- 326 4. *Ecosystem change*: such as change in ecosystem structure, function, integrity, ex-
327 tent, connectivity, or collapse risk.

328 These four forms operationalize the CBD definition at different levels of biological
329 organization in this systematic map. Genetic change corresponds to biodiversity *within*
330 *species*; species change and community change together capture important ways in which
331 biodiversity *between species* is reported in the literature; and ecosystem change captures
332 biodiversity *of ecosystems* [11]. The forms are not mutually exclusive: a single study may
333 report change at more than one level.

334 **Direction of change:** Biodiversity change may be negative (decline, degradation, loss),
335 positive (recovery, restoration, increase), mixed, or unclear. The analytical focus of this
336 systematic map is on biodiversity loss, that is, negative change in the state of biodiversity
337 at one or more of these levels. Records reporting other directions of change are retained
338 and coded but treated separately during synthesis.

339 **Direct anthropogenic drivers:** Biodiversity change must be linked to at least one
340 direct anthropogenic driver. The broad driver categories follow the IPBES conceptual
341 framework: land/sea-use change, direct exploitation and resource extraction, climate
342 change, pollution, and invasive alien species [2]. Specific human activities and pressures
343 within these categories are further characterized using the IUCN Threats Classifica-
344 tion Scheme [12], which provides a hierarchical vocabulary of proximate threats to
345 biodiversity.

346 **Linkage:** A record constitutes evidence when it reports or implies a link between one
347 or more direct anthropogenic drivers and the reported biodiversity change. This link may
348 take different forms, ranging from direct empirical observation to model-based projection,

349 and its nature is recorded as part of the screening output.

350 Together, these four components define what this systematic map treats as evidence:
351 a study that reports biodiversity change, describes its direction, identifies an anthropo-
352 genic driver, and connects the two. The Screening Strategy section describes how these
353 components are operationalized into inclusion and exclusion decisions.

354 **Supplementary Text S2**

355 **Prompts:** Prompts for screening and all label sets are provided in the data repository (see
356 Data Availability section). Each prompt specifies a constrained set of output labels; for the
357 majority of label sets, these are derived from established classification schemes: IPBES
358 Direct Drivers (L1), IUCN Threats Classification (L2), IPBES Geography (L3), Global
359 Ecosystem Typology (L4), and GBIF Taxonomy (L6). Label definitions and decision rules
360 are adopted from source materials; for instance, IUCN threat definitions and notes are
361 incorporated verbatim. The data repository provides the full label inventory with source
362 definitions for each label set. Prompts will be further refined iteratively on the training
363 split. Final configurations will be frozen prior to full-corpus application.

364 **Parameters:** Only reasoning effort will be varied during tuning. The reasoning effort
365 setting will be selected on the development split.

366 **Data Splits:** The validation datasets are partitioned into training (60%), development
367 (20%), and test (20%) splits using stratified sampling with a fixed random seed, stratified
368 by label column and original label sources. The training split was used for initial prompt
369 development and will be used for further prompt refinement; the development split will be
370 used for model selection and reasoning effort tuning; the test split will be held out for final
371 consistency checking prior to full-corpus application.

372 **Freezing Condition:** Prompts and reasoning effort will be refined iteratively. After
373 each iteration, consistency checking metrics described in the Consistency Checking sec-
374 tions will be evaluated on training and development splits. Configurations will be frozen
375 when no significant improvement in these metrics is observed across iterations. Prior to

³⁷⁶ full-corpus application, a final evaluation on the held-out test split will be conducted to
³⁷⁷ confirm consistency checking results.